	GLOBAL ECONOMICS **GRADE: 10** **LEARNING EVIDENCE 2** **DECISION-MAKING UNDER UNCERTAINTY** **TEACHER'S NAME: Nicolás López Cuéllar**	**1ST TERM** **2026–2027**

Learning objective: Apply minimax regret and Expected Monetary Value (EMV) to select an alternative under uncertainty, and use sensitivity analysis to determine how a subjective probability changes the preferred alternative.	**Assessment criteria:** C5: Constructs opportunity loss tables and applies the Minimax regret criterion to select the alternative that minimises the maximum opportunity loss. C6: Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff. C7: Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

Each problem is worth 2 points. Show all calculations, including the complete regret table for C5, every EMV calculation for C6, and the threshold equation and decision ranges for C7. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. No points will be awarded if no procedure is shown. A **minimum of 4 points** is required to pass each criterion. A 1-point penalty will be applied to each criterion if the student's name is not written on the assessment or if the assessment is submitted late.

Question	C5	C6	C7
1	2	–	–
2	2	–	–
3	2	–	–
4	–	2	–
5	–	2	–
6	–	2	–
7	–	–	2
8	–	–	2
9	–	–	2
Total	**6**	**6**	**6**

1. Artisan Apiary Expansion [C5-2]

An artisan apiary must choose one of three hive-expansion plans. Annual net benefits are in thousands of dollars.

	Alternatives		
	Large	Medium	Small
Excellent	76	58	38
Average	34	49	43
Poor	-14	23	35

What decision would be selected using the Minimax Regret criterion? Justify your answer.

2. Regional Theater Capacity [C5-2]

A theater operator is choosing a seating-capacity plan. Seasonal profit is in thousands of dollars; no reliable probabilities are available.

Alternatives	High	Normal	Low
Add two seating sections	108	40	-28
Add one seating section	82	61	14
Lease a nearby hall	59	52	34
Current seating	34	34	34

What decision would be selected using the Minimax Regret criterion? Justify your answer.

3. Emergency Shelter Staffing [C5-2]

A relief organization must select an emergency-shelter staffing plan. Net community value is in thousands of dollars.

	Alternatives		
	Full	Flexible	Limited
Very high	96	74	48
High	54	64	51
Moderate	24	49	46
Low	-24	12	38

What decision would be selected using the Minimax Regret criterion? Justify your answer.

4. Coffee Roastery Product Mix [C6-2]

A coffee roastery is selecting a product mix. Profits are in thousands of dollars.

		Strong	Stable	Weak
A_i	$P(S_j)$	0.45	0.35	0.20
Single-origin		86	44	-12
Balanced		64	58	22
Classic		42	47	37

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

5. Aquarium Program Scale [C6-2]

An aquarium is deciding the scale of a temporary education program. Payoffs are in thousands of dollars.

		High	Expected	Low
A_i	$P(S_j)$	0.30	0.40	0.30
Major		98	38	-24
Standard		73	53	18
Compact		52	47	34
Regular tours only		27	27	27

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

6. Wind-Farm Maintenance Staffing [C6-2]

A wind-farm operator is choosing a maintenance staffing plan. Payoffs are in thousands of dollars.

		Surge	Strong	Steady	Slow
A_i	$P(S_j)$	0.20	0.30	0.35	0.15
Rapid hiring		105	64	24	-32
Moderate hiring		79	68	48	12
Current staff		53	55	51	42

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

7. Concert Merchandise Order [C7-2]

A concert vendor must choose between a large and a small merchandise order. Payoffs are in thousands of dollars.

A_i	High (S_1)	Low (S_2)
Large	80	20
Small	50	40

Let $P(S_1) = p$. State the preferred alternative for every probability value ($0 \le p \le 1$).

8. Meal-Kit Launch [C7-2]

A meal-kit company is comparing a bold launch with a measured launch. Payoffs are in thousands of dollars.

	High	Moderate	Low
A_i	S_1	S_2	S_3
Bold	60	30	0
Measured	35	45	30

Suppose $P(S_1) = p$, $P(S_2) = \frac{p}{3}$. State the preferred alternative for every feasible probability value of p.

9. Publishing-Platform Bid [C7-2]

A publishing firm must choose between a standard platform bid and an experimental platform bid. Payoffs are in thousands of dollars.

	Immediate	Revision	Delayed
A_i	S_1	S_2	S_3
Standard	30	70	70
Experimental	50	50	30

Suppose $P(S_1) = 3p$, $P(S_2) = p$. State the preferred alternative for every feasible probability value of p.